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I. Overview

Chapter 1

Combustion Instabilities: Basic Concepts

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I. Introduction

COMBUSTION instabilities are characterized by large-amplitude oscillations of one or more natural acoustic modes of the combustor. Such instabilities have been encountered during the development and operation of propulsion (e.g., rockets, ramjets, and afterburners), power generation (e.g., land-based gas turbines), boiler and heating systems, and industrial furnaces. These instabilities are spontaneously excited by a feedback loop between an oscillatory combustion process and, in general, one of the natural acoustic modes of the combustor. In general, the occurrence of instabilities is problematic because they produce large-amplitude pressure and velocity oscillations that result in thrust oscillations, severe vibrations that interfere with control-system operation, enhanced heat transfer and thermal stresses to combustor walls, oscillatory mechanical loads that result in low- or high-cycle fatigue of system components, and flame blowoff or flashback. These phenomena may result in premature component wear that could lead to costly shutdown or catastrophic component and/or mission failure. Consequently, considerable research and development efforts have been invested during the past half-century to elucidate the processes responsible for the excitation of these instabilities and the development of approaches for their prevention. The objective of this chapter is to provide an overview of the causes, characteristics, and control of these instabilities.

Figure 1.1 summarizes the conditions under which combustion instability spontaneously occurs. The cartoon on the top shows an unstable combustor with reactants entering on the left and combustion products leaving through the nozzle on the right. Interaction between one of the combustor's acoustic modes and heat-release oscillations transfers or removes energy from the acoustic mode. It can

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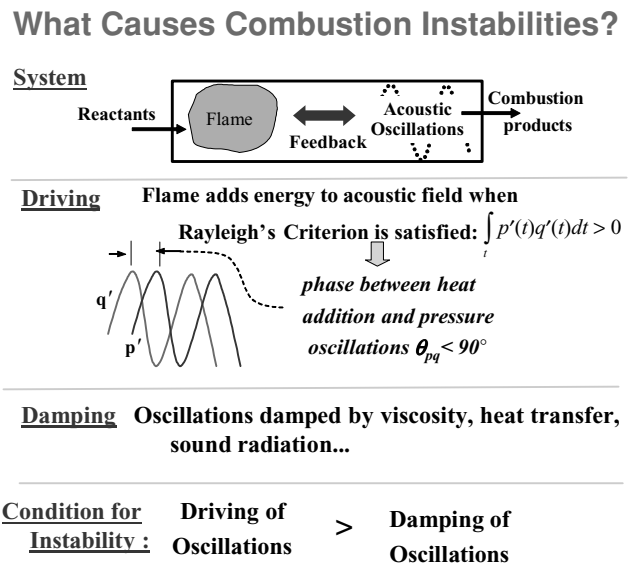

Fig. 1.1 Summary of conditions required for a combustion instability to occur.

be shown that the combustion process adds (removes) energy from the acoustic oscillations locally if the integral in Fig. 1.1, which is often referred to as Rayleigh's integral,[1] is positive (negative). The sign of this integral depends on the phase difference between the heat-release and pressure oscillations and is positive (negative) when this phase difference is smaller (larger) than 90 deg. As shown in Fig. 1.1, combustion instability spontaneously occurs only if the energy supplied to the acoustic mode by the combustion process exceeds the energy losses of the mode caused by, for example, radiation and convection of acoustic energy out of the combustor (e.g., through the nozzle), viscous dissipation, and heat transfer. Thus, as long as the magnitude of the driving exceeds the magnitude of the damping process, the energy of the mode will increase with time. In such a case, the amplitude of oscillations initially increases exponentially with time until it saturates at some limit-cycle amplitude. When this occurs, the time averages of the driving and damping processes are equal and no net energy is added to the oscillating mode.

In general, combustion instabilities occur at frequencies associated with natural acoustic modes of the combustor. These include, for example, bulk (i.e., Helmholtz-type oscillations), axial, and transverse (i.e., tangential and/or radial) modes (see Fig. 1.2). On occasion, however, the oscillations are not associated with a purely acoustic mode and are excited by a coupled convective-acoustic mode similar to cavity tones,[2] which often occur at frequencies lower than those of purely acoustic modes. Such oscillations occur when an entropy wave (i.e., a hot-gas packet) or a vortex generated in the flame region is convected toward (and impinges on) the nozzle, at which it excites an acoustic wave that propagates back to the flame,[3,4] exciting another convected wave and thus repeating the process. These types of modes are often encountered in systems that are operating at conditions close to flame blowoff.

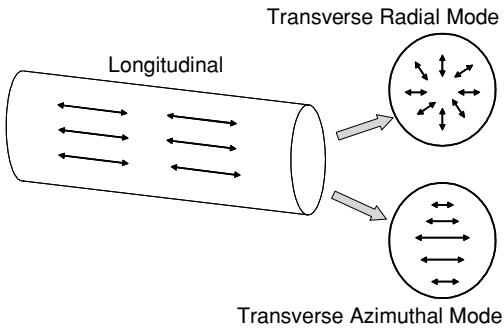

Fig. 1.2 Examples of longitudinal and transverse acoustic modes that are excited in cylindrical combustors.

Because the initial amplitudes of most instabilities are generally quite small, their characteristics are described by the linear-wave equation.[5] The frequencies and mode shapes of these oscillations and the conditions under which they spontaneously occur are determined by the solutions of these equations. Linear analyses cannot, however, predict the magnitude of the limit-cycle amplitude attained by the instability because it is controlled by nonlinear processes. Furthermore, nonlinear processes may allow a large-amplitude disturbance whose amplitude exceeds a certain threshold value, A_T, to trigger instability in a system that is linearly stable; that is, one in which low-amplitude oscillations are not spontaneously self-excited. Consequently, both the characteristics of the limit-cycle oscillations and conditions under which finite amplitude disturbances trigger instabilities can only be determined by solving the nonlinear equations that describe the system's dynamics.[6,7]

To prevent the onset of detrimental combustion instabilities, the processes responsible for their driving and damping must be understood. This chapter presents an overview of the current understanding of these processes with the objective of providing the reader with the background needed for the more detailed discussions of these and related subjects in the remaining chapters of this volume. To attain this goal, the following topics are discussed in this chapter: the history of this problem, driving and damping of combustion instabilities, common instability mechanisms in gas turbines, the initial growth and saturation of the oscillations, and some basic characteristics of limit-cycle oscillations.

II. Historical Overview

This section provides a brief overview of prior experience with combustion instabilities. Although instabilities have been observed in a variety of combustion systems, related phenomena have also been excited in other systems by heat-transfer processes.[8] For example, glass blowers have reported observations of spontaneous excitation of acoustic oscillations during the heating of the closed end of a blown-glass tube. Also, acoustic oscillations can be spontaneously excited inside tubes with sharp-temperature gradients, as has been observed in cryogenic systems.[9] Because in all these cases the acoustic oscillations are excited by thermal sources, the resulting phenomena are often referred to as thermoacoustic instabilities.

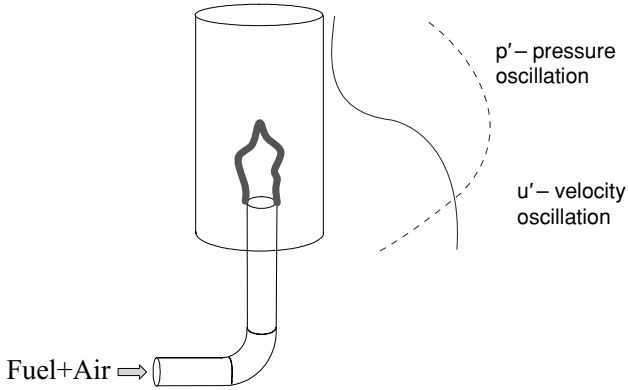

Fig. 1.3 An example of a flame-driven instability in a tube open at both ends.

With a focus on combustion-driven oscillations for the remainder of the chapter, the first observation of flame-driven oscillations (referred to as singing flames) dates back to 1777.[10] This and subsequent studies found that spontaneous acoustic oscillations of considerable amplitude might be generated when a gas flame is placed inside a larger-diameter tube, as illustrated in Fig. 1.3. The sensitivity of flames to music at a musical party was noted by Le Conte[11] in 1858 and described in the following quotation: "Soon after the music commenced, I observed that the flame exhibited pulsations exactly synchronous with the audible beats. This phenomenon was very striking to everyone in the room, and especially so when the strong notes of the violoncello came in.... A deaf man might have seen the harmony...."

Combustion oscillations moved beyond an academic curiosity with the advent of high-intensity combustion systems. Detrimental, combustion-driven oscillations have been observed in boilers, blast furnaces, and a variety of other oil, coal, and gas-fired heating units.[12] The occurrence of these instabilities was generally unexpected as demonstrated, for example, by recent experience with one of the nation's largest landfill sites in Los Angeles County (Fig. 1.4). The site has two 45-ft-high, 12-ft-diam flares, each designed to flare excess landfill gas at rates of up to ~50 MW. However, at loads greater than about 50% of full capacity, instability of the quarter-wave mode of the flare, oscillating at about 10 Hz, is excited in the system.[13]

Instabilities have also significantly hindered the development of various liquid-fueled rockets. Notable were the instabilities encountered during the development of the F-1 engine that powered the Saturn rockets, which were used in the first manned mission to the moon. The F-1 encountered instabilities with amplitudes up to 100% of the mean combustor pressure (i.e., more than 2000 psi) with frequencies in the 200- to 500-Hz range. These instabilities caused significant damage to the combustor, and their elimination required a costly trial and error, development program that included ~2000 full-scale tests (of a total of 3200). One of the solutions developed involved welding a system of baffles to the injector face. These baffles prevented the excitation of the transverse acoustic oscillations that could be driven by the combustion process near the injector face (Fig. 1.5).

Fig. 1.4 Landfill gas flares that experienced low-frequency, combustion-driven oscillations.[13]

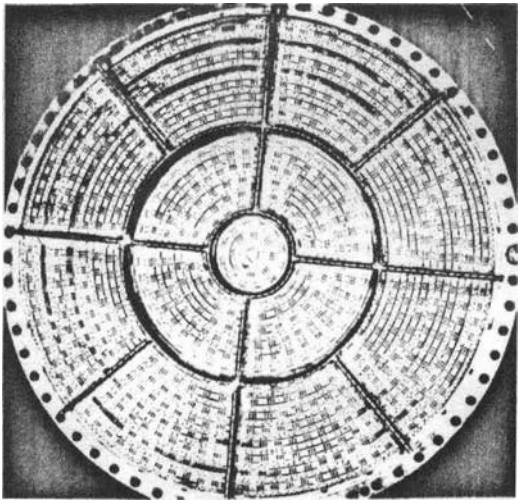

Fig. 1.5 Photograph of rocket injector with welded baffle plates that was tested to prevent transverse mode instabilities during F-1 engine development.[26]

Combustion instabilities have also been encountered in numerous solid-propellant rockets, including the Space Shuttle solid-propellant rocket boosters, the Minuteman intercontinental ballistic missile, and the Mars Pathfinder descent motor.[14] The Minuteman missile provides a good example of the enormous difficulties that instabilities generated in solid-propellant rocket development and testing programs. In 1968, the U.S. Air Force experienced five flight failures of the Minuteman Wing I missiles during routine tests. This rate of failure raised concerns about the condition of the systems that were already fielded in missile silos and resulted in costly removal and modifications of many of the systems.

Combustion instabilities in ramjet-powered missiles have also been problematic because they cause thrust oscillations and shock-system oscillations in the inlet diffuser, which lead to a reduced stability margin of the inlet flow.[15] Similar problems have also been encountered in afterburners, in which instabilities of transverse acoustic modes, generally referred to as "screech," and axial modes damage flame holders, liner sections, and other engine components.[15]

III. Causes of Instabilities

A. Combustion Process: Acoustic Modes Energy Transfer

This section discusses the mechanisms through which the combustion process can drive acoustic modes of the system. Lord Rayleigh[1] was the first to state the so-called Rayleigh criterion, which describes the conditions under which a periodic heat-addition process adds energy to acoustic oscillations. Paraphrased, the criterion states that a periodic heat-transfer process adds energy to the acoustic field if the heat is added to or removed from the gas when its pressure is above or below its mean value. This statement is described mathematically by the integral in Fig. 1.1. It basically says that the heat-addition process locally adds energy to the acoustic field when the magnitude of the phase between the pressure and heat-release oscillations, θ_{pq}, is less than 90 deg (i.e., $0 < |\theta_{pq}| < 90$). Conversely, when these oscillations are out of phase (i.e., $90 < |\theta_{pq}| < 180$), the heat-addition oscillations damp the acoustic field.

The physical reason for this energy exchange follows from determining the conditions under which the unsteady heat release performs work on the gas. Heat release at constant pressure results in gas expansion, analogous to blowing up a balloon. The Rayleigh criterion states that the unsteady heat release performs work on the gas when this expansion occurs in phase with the pressure. This statement is analogous to problems in mechanics in which an unsteady force (i.e., the pressure) must be in phase with the velocity (i.e., the gas-dilatation rate) if net work is to be performed.

One may also gain physical insight into the Rayleigh criterion by considering its similarities to the manner in which the combustion-process energy is converted into work in a Brayton cycle, the thermodynamic cycle used in gas turbines. In this cycle, thermal energy is added to the working fluid at high pressure after it has been compressed in the diffuser and compressor. Work is subsequently extracted from the working fluid at low pressure as it is expanded through the turbine and nozzle. The similarity between the Brayton cycle and the Rayleigh criterion becomes apparent if one recalls that according to the Rayleigh criterion one can drive acoustic pressure oscillations in a system by heating the gas with

an instantaneous "heat-addition pulse" when its pressure is maximum and cooling it with an instantaneous "cooling pulse" when its pressure is minimum. Such a process essentially describes the manner in which heat is added and removed in a Brayton cycle to do work.

The Rayleigh criterion describes the conditions under which unsteady heat release adds energy to the acoustic field. However, even if energy is transferred from the combustion process to the acoustic field, this does not necessarily imply that the combustor is unstable. As described in Fig. 1.1, acoustic oscillations are spontaneously excited in a combustor only when the rate of energy supplied by the periodic combustion process to the acoustic field is larger than the rate at which acoustic energy is dissipated within the combustor and/or transmitted through its boundaries. This statement is summarized by the following expression:

$$\iint_{V\,T} p'(x,t)\,q'(x,t)\,\mathrm{d}t\,\mathrm{d}V \geq \iint_{V\,T} \sum_i L_i(x,t)\,\mathrm{d}t\,\mathrm{d}V \tag{1.1}$$

where $p'(x,t)$, $q'(x,t)$, V, T, and L_i are the combustor pressure oscillations, heat-addition oscillations, combustor volume, period of the oscillations, and the ith acoustic energy loss process (e.g., viscous dissipation, radiation of acoustic energy out of the combustor through its boundaries), respectively. The equal sign in Eq. (1.1) describes conditions when limit-cycle oscillations are attained and the time average of the energies added and removed from the oscillations are equal. The integral on the left side of Eq. (1.1) is referred to as the Rayleigh integral and is often used in experimental or numerical studies to quantify the energy transfer from the combustion process to the acoustic field. Note that the inner integral on the left side of Eq. (1.1) is the integral shown in Fig. 1.1, which describes the *local* driving/damping of the acoustic oscillations by the heat-addition process $q'(x,t)$.

B. Instability-Driving Mechanisms

As discussed earlier, combustion instabilities are excited by feedback between the combustion process and acoustic oscillations that depends on the system characteristics and operating conditions. This section provides a brief overview of common instability mechanisms and the conditions under which they are self-exciting.

Figure 1.6 describes the generic feedback loop responsible for combustion instabilities. It consists of the following sequence of events: 1) Fluctuations in the velocity and/or thermodynamic-state variables induces a fluctuation in the heat-release rate, 2) the heat-release fluctuation excites acoustic oscillations, and 3) the acoustic oscillations generate the velocity and thermodynamic-state variable fluctuations that are described in step 1, thus closing the feedback loop. Depending on the relative magnitudes of the energy added and removed from the acoustic oscillations, the amplitude of oscillations may decrease, remain constant, or grow during each cycle of this loop.

Several mechanisms capable of driving combustion instabilities in gas turbines have been identified, as indicated in Fig. 1.7. Because most of these mechanisms

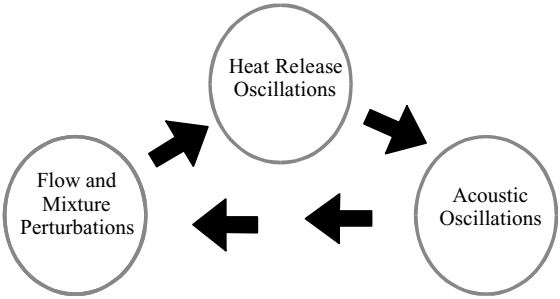

Fig. 1.6 Illustration of the feedback processes responsible for combustion instability.

are discussed in more detail in Chapter 9, the remainder of this section only provides a brief description of each.

1) Fuel Feed Line–Acoustic Coupling.[5,16] Pressure oscillations in the combustor modulate the pressure drop across unchoked fuel nozzles. The pressure drop, in turn, modulates the fuel-injection rate into the system, causing an oscillatory heat-release process that drives the acoustic oscillations.

2) Equivalence-Ratio Oscillations.[17] Combustor pressure oscillations propagate into the premixer section in which they modulate mixing processes and fuel and/or air supply rates, thus producing a reactive mixture whose equivalence ratio varies periodically in time. The resulting mixture is convected into the flame in which it produces heat-release oscillations that drive the instability.

3) Oscillatory Atomization, Vaporization, and Mixing.[5,16] Interactions of the acoustic field with the fuel spray produce periodic variations of the fuel-spray shape, droplet sizes, evaporation rates, and mixing rates of the fuel vapor with surrounding gases. These variations, in turn, could result in periodic supply rates of fuel to the flame and/or periodic variations of the equivalence ratio that produce heat-release oscillations that drive the acoustic field.

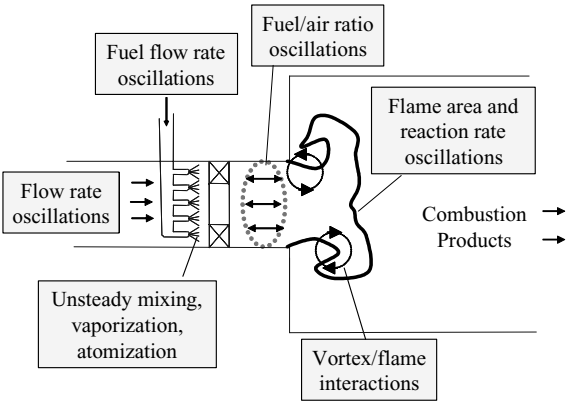

Fig. 1.7 Flow and flame processes that can cause combustion instabilities in gas turbines.

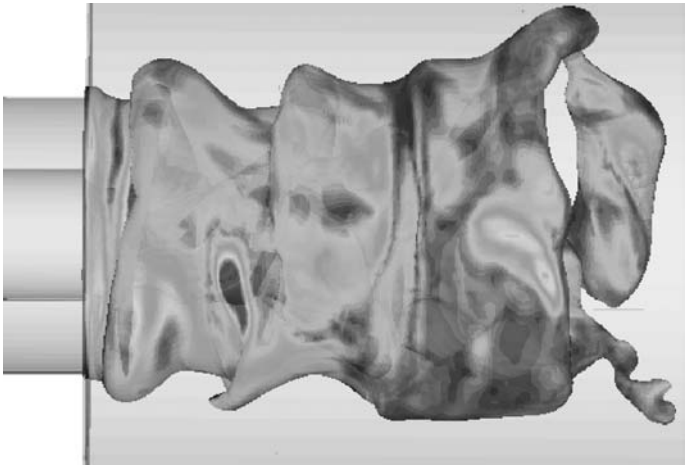

Fig. 1.8 Computed image of swirling flame distorted by vortical structures. Courtesy of Y. Huang and V. Yang.[40]

4) Oscillatory Flame-Area Variation.[18] Interactions of acoustic velocity oscillations with the flame cause periodic variation of the flame area and, thus, a periodic heat-addition process that drives the acoustic field.

5) Vortex Shedding.[19,20] Large-scale, coherent vortical structures caused by flow separation from flame holders and rapid expansions, as well as vortex breakdown in swirling flows, are often present in gas-turbine combustors, as shown in Fig. 1.8. In the initial stage of their formation, these vortices generally consist of combustible gases. As they form, these vortices entrain hot products and ignite. This ignition is followed by rapid combustion of the reactants within the vortex and sudden breakdown of the coherent vortical structure into small-scale turbulence. Alternatively, the vortical structures may distort the flame and cause its surface area to oscillate, thus producing an oscillatory heat-release rate process that can also drive the acoustic field if the heat addition and pressure oscillations are in phase. These dynamics are also discussed in Chapters 10 and 11 of this volume.

Heat-release oscillations add energy to an acoustic field if the magnitude of the characteristic timescales of the heat-addition process are of the order of some integer multiple of the acoustic period. This point is illustrated in Fig. 1.9, which shows the hypothetical response of 1) the fuel-injection rate, 2) the rate of change of droplet-size distribution, 3) the rate of heat transfer to the fuel, and 4) the rate of fuel–air mixing to a pressure pulse with period T. The fluctuations in each of the rates of the preceding four processes causes a fluctuation in the heat-release rate after a time delay τ, which depends on the characteristics of the analyzed process, the combustor-operating conditions, and design (Fig. 1.9). For example, the magnitude of the time delay τ_1 depends on various geometric and operating parameters of the fuel-delivery system; for example, the fuel-line length and fuel gas temperature.

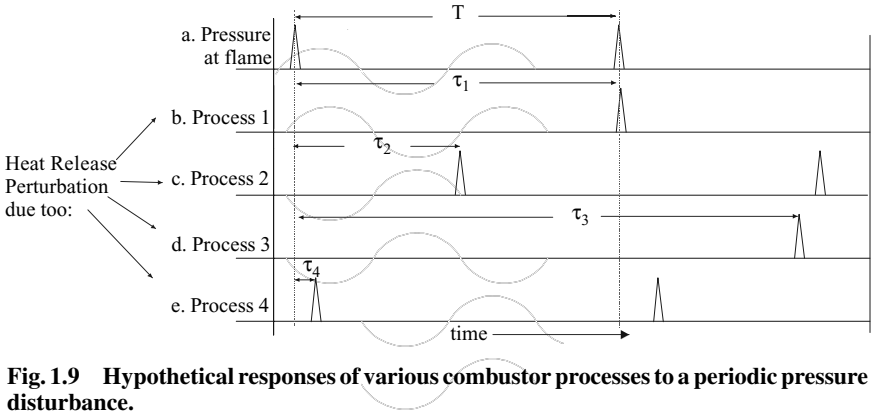

Fig. 1.9 Hypothetical responses of various combustor processes to a periodic pressure disturbance.

According to the Rayleigh criterion, the heat-release oscillations shown in Fig. 1.9 add energy to the acoustic field if the time average of the product of the pressure and heat-release oscillations is greater than zero. Figure 1.9 shows that the heat release attributable to process 1 satisfies this criterion and that the heat release by processes 2–4 does not. The heat release of process 1 and the pressure pulses are positively correlated because the characteristic time of the process τ_1 is of the order of the period of the acoustic waves, T. Combustion instability will thus occur if the rate of energy addition to the disturbance by process 1 exceeds its rate of damping.

Keep in mind that any of a number of the natural acoustic modes of the combustion system can be excited. To illustrate the relationships between different acoustic modes and various driving mechanisms, assume that the pressure disturbance of period T in Fig. 1.9 is caused by oscillations of the first longitudinal acoustic mode of the combustor. If we now repeat the preceding analysis to determine the driving of the second longitudinal acoustic mode of the combustor with a period of $T/2$, then $\tau_1 = 2T$ and $\tau_2 = T$. Consequently, in the example shown in Fig. 1.9, energy is added to this mode by both processes 1 and 2. This example shows that an instability could be excited if the characteristic time of the combustion process equals T, $2T$, or any other integer multiple of T.

The preceding examples illustrate several important points. First, they show that different mechanisms may play different roles in the stability of different modes of the combustor. For example, the mechanism(s) responsible for exciting a 100-Hz longitudinal mode of a combustor will significantly differ from those that excite a 5000-Hz transverse-mode instability. Second, these examples show that the role of various mechanisms may change with operating conditions. For example, suppose changes in combustor-operating conditions change the timescales of processes 1 and 2. Such a change in timescales could affect the coupling between the pressure and heat-addition disturbances and, thus, the role of each process in driving various combustor modes. Finally, the preceding examples demonstrate that the characteristic times associated with processes that are responsible for the excitation of

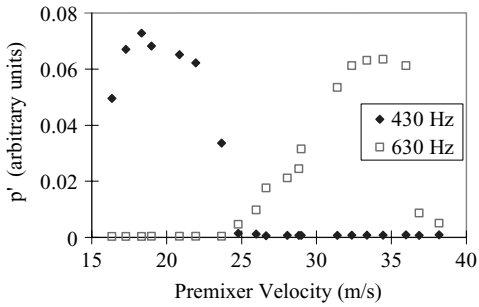

Fig. 1.10 Measured[25] dependence of the instability amplitude on the premixer velocity.

combustion instability must have magnitudes that are of the order of some integer multiple of the acoustic period of the modes.

Note also that combustor pressure oscillations generally vary harmonically with time and do not exhibit the pulselike behavior used in the examples in Fig. 1.9. Consequently, the time delays and acoustic periods of the various modes need not be exactly equal for the process to add energy to the acoustic field; instead, these characteristic times must satisfy a relationship of the form $T - T/4 < n\tau < T + T/4$, where n is an integer; that is, $n = 1, 2, \ldots$[17]

Some of these points are illustrated by the results in Fig. 1.10,[17] which describe the dependence of the amplitudes of a 430- and 630-Hz combustor mode on the mean velocity of the reactants in the combustor premixer. The instabilities in this combustor were driven by a mechanism that depends on the time required to transport the reactants from the fuel-injection point to the combustor,[17] a time delay that is inversely proportional to the mean velocity of the gases in the premixer. Consequently, this time delay decreases as the gas velocity in the premixer increases. Figure 1.10 shows that, as the velocity in the combustor premixer increased and the convective time delay decreased, the amplitude of the lower-frequency mode (i.e., 430 Hz with the longer period T) decreased and the amplitude of the higher-frequency mode (i.e., 630 Hz with the shorter period T) increased. This finding indicates that when the premixer velocity was low the driving process coupled with the 430-Hz mode and this coupling "switched" to the 630-Hz mode as the premixer velocity increased.

C. Damping Processes

As noted earlier, acoustic damping processes play an important role in the determination of the conditions under which combustion instability occurs. This section summarizes important damping mechanisms in combustors and the key parameters that influence their magnitude.

Energy can be dissipated/removed from an unstable mode via the following three processes: 1) transfer of acoustic energy to vortical or entropy disturbances through viscous and heat-transfer processes, respectively; 2) convection and/or radiation of acoustic energy out of the system; and 3) transfer of energy between acoustic

modes. Note that in all these cases, the "dissipation process" refers to the transfer of acoustic energy out of the combustor, frequency regions, or modes of oscillation in a manner that reduces the acoustic energy of the unstable mode.For example, in the second mechanism, the acoustic energy is simply radiated or convected out of the combustor through its boundaries and represents part of the noise heard outside the system. The rest of this section describes each of these mechanisms in further detail.

1. Viscous and Heat-Transfer Damping Mechanisms

This mechanism can be further subdivided into two submechanisms: boundary-layer losses and flow-separation losses. Boundary-layer losses occur when acoustic motions are present in the vicinity of surfaces on which viscosity and thermal-dissipation effects dominate (see Ref. 21). Suppose that an acoustic wave that perturbs the flow velocity and temperature impinges obliquely on a rigid wall. Because of the no-slip boundary condition at the wall, the energy in the acoustic mode is partially transferred into vortical velocity fluctuations. In an analogous manner, the temperature-boundary condition at the wall (e.g., zero-amplitude temperature oscillations) causes some acoustic energy to be converted into entropy fluctuations. Consequently, the energy of the acoustic wave reflecting from the wall is smaller than that of the incident wave, because some of its energy is converted into vorticity and/or entropy fluctuations. These processes are analogous to those occurring in a steady pipe flow in which heat transfer and viscosity result in stagnation-pressure losses. The magnitudes of these dissipation mechanisms increase with frequency as $(f\tau_v)^{1/2}$, where τ_v is the viscous or thermal transport timescale.

Flow separation at sharp edges or during rapid flow expansions also damp acoustic waves by converting acoustic energy into vorticity. This damping mechanism is analogous to the stagnation-pressure losses in steady, separated flow downstream of sharp corners or edges, tabulated in many engineering handbooks and fluid mechanics texts.[22] This dissipation mechanism has nonlinear-amplitude dependence and is discussed further in Sec. IV.C.

2. Convection and/or Radiation of Acoustic Energy

Acoustic energy inside a duct can leave the system by propagation and/or convection by the mean fluid motion out of the system. For example, the tone heard from an organ pipe is caused by some of the sound energy in the standing wave inside the pipe propagating out through its open end. In general, this damping mechanism scales with frequency as $(fD/c)^2$, where f, D, and c are the frequency, pipe diameter, and sound speed, respectively. Sound energy is also convected out of the system by fluid motion with a magnitude that roughly scales with the mean flow Mach number. Because the magnitude of acoustic damping in a typical system is quite small, in general, the presence of low-velocity mean flow can have a significant impact on the system's damping level. To illustrate this point, consider the dependence of the magnitude of a reflected acoustic wave incident on the open end of a pipe upon the flow Mach number in Fig. 1.11. It shows that the reflection coefficient equals 0.95 in the no-flow case, implying that the wave amplitude is reduced by 5% because of acoustic radiation. This reflection coefficient is reduced to

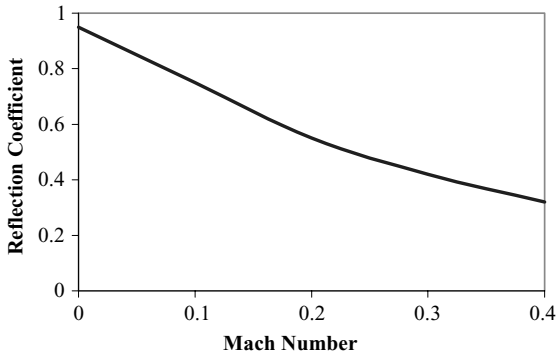

Fig. 1.11 Dependence of the reflection coefficient from an open-ended pipe on the flow Mach number. Taken from Ingard and Singhal.[41]

0.85 at $M = 0.05$, implying that the presence of a very-low-velocity flow increases the acoustic damping by a factor of 3.

3. *Transfer of Energy Out of Natural Acoustic Frequencies to Other Frequencies*

Oscillations in an unstable combustor, in general, are composed of one or more nearly pure tones at specific frequencies. This composition occurs because combustors are generally lightly damped acoustic resonators that respond to external disturbances over very-narrow-frequency ranges. Consequently, mechanisms that do not directly dissipate acoustic energy but transfer energy from the excited modes to other modes that oscillate at frequencies that are either not amplified in the combustor or at which the energy is more readily dissipated, effectively constitute "dissipation" mechanisms for a given natural acoustic mode. The energy transfer can be to either narrowband, coherent, fluctuations at other frequencies or to incoherent, broadband fluctuations. Nonlinear combustor processes are a key mechanism for enabling the former; that is, they transfer energy from a certain frequency f_o, to higher harmonics ($2f_o$, $3f_o$, ...) or subharmonics ($f_o/2$, $f_o/3$, ...). The energy at these higher frequencies is usually dissipated more rapidly by the previously discussed viscous and radiation mechanisms whose magnitudes increase with frequency. Nonlinear combustor processes are discussed further in Sec. IV.

Acoustic energy is transferred from narrowband, coherent oscillations to spectrally distributed, incoherent oscillations by random modulation processes. For example, such spectral broadening occurs during reflection and scattering of a sound wave from turbulent eddies[23] or a randomly flapping flame front.[24] The energy-transfer mechanism in this case can be purely linear and is essentially caused by a random Doppler shift of the reflected and transmitted waves. For example, when a pure tone with a temporal dependence given by $\sin(\omega t)$ impinges on a randomly moving flame front or a medium with a random velocity field, reflected and transmitted waves with a random phase shift are generated. The temporal dependence of the reflected/transmitted waves is given by $\sin[\omega t + \phi(t)]$, where $\phi(t)$ is a random phase shift. This mechanism is discussed further in Chapter 12.

IV. Growth and Saturation of Instabilities

As indicated by Eq. (1.1), the amplitude of the instability grows if the rate of energy addition to the oscillations exceeds the rate of energy dissipation by damping processes. As the amplitude of the oscillations increases, the energy addition and dissipation processes become amplitude dependent and the amplitude of the oscillations attains its maximum value when the time average of the energy addition and removal equal one another. The resulting oscillations are referred to as a limit cycle. The objective of this section is to consider in more detail the growth and saturation of the instability amplitude.

A. Linear and Nonlinear Stability

As discussed in the introduction, the mechanisms that initiate combustion instabilities are typically grouped into linear and nonlinear categories. A linearly unstable system is one that is unstable with respect to infinitesimally small disturbances. Thus, because linearly unstable systems can not exist, they are never observed in nature. An example of a linearly unstable situation is a ball perfectly balanced at the crest of a hill, where any small disturbance will cause it to roll away from this unstable equilibrium point.

To further illustrate the dependence of the limit cycle on the amplitude of the oscillations A consider the hypothetical, amplitude-dependent, driving, $\mathrm{H}(A)$, and damping, $\mathrm{D}(A)$ processes, which are described in Fig. 1.12 and the following expressions:

$$\mathrm{H}(A) = \varepsilon_{\mathrm{H}} A + \mathrm{H}_n(A) \tag{1.2}$$

$$\mathrm{D}(A) = \varepsilon_{\mathrm{D}} A + \mathrm{D}_n(A)$$

where subscript n denotes the nonlinear part of the transfer function.

As shown in Fig. 1.12, the driving and damping curves intersect at the origin, indicating that a zero-amplitude oscillation is a potential equilibrium point. This equilibrium point is unstable, however, because any small disturbance that moves the system away from the origin produces a condition in which $\mathrm{H}(A)$ is larger than

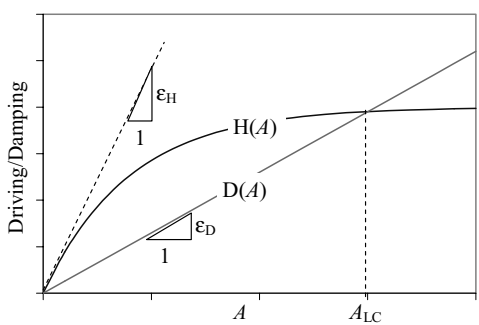

Fig. 1.12 Hypothetical dependence of the acoustic driving, $\mathrm{H}(A)$, and damping, $\mathrm{D}(A)$, processes on the instability amplitude A.

D(A), resulting in further growth of the disturbance. Because these two curves diverge near the origin, their difference increases with amplitude, implying that the amplitude growth rate increases with amplitude. When the amplitudes of the oscillations are small, the driving and damping processes can be approximated by linearized expressions; that is, $\mathrm{H}(A) = \varepsilon_{\mathrm{H}} A$ and $\mathrm{D}(A) = \varepsilon_{\mathrm{D}} A$. Thus, linear combustor processes (i.e., processes whose magnitude is directly proportional to amplitude A), in general, control the balance between driving and damping processes when the amplitude of oscillations is small and thus determine the frequency and growth rate $A \sim e^{\alpha t}$ of inherent combustor disturbances. It can be shown that the initial growth rate of the instability α is proportional to the difference between the driving and damping processes in the linear regime; that is, $\alpha \sim \varepsilon_{\mathrm{H}} - \varepsilon_{\mathrm{D}}$. For this reason, linear combustor stability models (such as those described in Chap. 13) are routinely used to determine whether a given operating point is stable or unstable.

Nonlinear combustor processes control the dynamics of the oscillations as the driving and damping processes become amplitude dependent. Figure 1.12 describes a situation in which H(A) saturates and D(A) increases linearly with the amplitude A, thus resulting in an intersection of the two curves at the limit-cycle amplitude A_{LC}. Note that this limit-cycle amplitude is stable; that is, a perturbation of the amplitude to the left (right) of this intersection point causes H(A) to become larger (smaller) than D(A), thus causing the amplitude to increase (decrease) and return to its limit-cycle value A_{LC}.

Next, consider a situation in which some combustor parameter is systematically varied in such a way that ε_h increases while ε_d remains constant (see Fig. 1.12). For $\varepsilon_h < \varepsilon_d$, $A = 0$ is the stable solution; that is, the system is linearly stable because all perturbations imposed on the system will decay. However, when $\varepsilon_h > \varepsilon_d$, the solution $A = 0$ becomes unstable, causing the amplitude of the disturbance to increase toward a new stable, limit-cycle equilibrium. The $\varepsilon_h = \varepsilon_d$ condition separates two regions of fundamentally different dynamics and is referred to as a *supercritical bifurcation point*. These ideas are illustrated in the bifurcation diagram in Fig. 1.13, which shows the dependence of the amplitude A_{LC} on $\varepsilon_h - \varepsilon_d$. Figure 1.13 shows that as $\varepsilon_h - \varepsilon_d$ becomes positive, the system becomes linearly

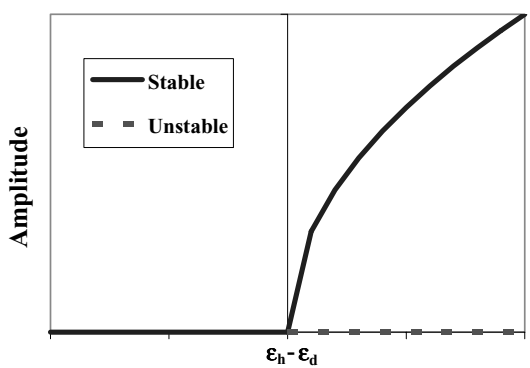

Fig. 1.13 An example of a supercritical bifurcation.

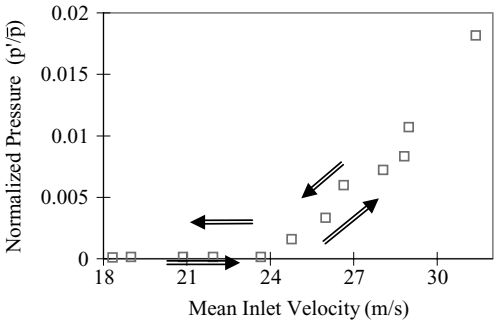

Fig. 1.14 Measured data describing the occurrence of supercritical bifurcation at combustion inlet velocity of 23.5 m/s.[25]

unstable. Although $\varepsilon_h - \varepsilon_d$ describes variation along the x-axis in Fig. 1.13, it could be replaced in practice by any parameter that affects the system's stability, for example, the air velocity or temperature as demonstrated by the measured data in Fig. 1.14.[25] It shows a smooth, monotonic dependence of the amplitude on the premixer velocity, which is indicative of the presence of the previously discussed supercritical bifurcation in this combustor.

A nonlinearly unstable system differs from a linearly stable one in that it is stable with respect to small-amplitude disturbances but is unstable when subjected to disturbances whose magnitude exceeds a certain threshold value A_T. This type of instability is sometimes referred to as *subcritical*. A simple example of a nonlinearly unstable system is shown in Fig. 1.15, which shows a ball in a depression on the top of a hill. When pushed, this ball returns to its equilibrium point as long as it is subjected to disturbances with amplitude that does not get it over the sidewalls of the depression. However, for a sufficiently large disturbance amplitude, the ball will roll out of the depression and down the hill. As discussed earlier, such behavior is often referred to as "triggering" in the context of combustion instabilities in rockets. A simple example of a nonlinearly unstable combustion system is one in which the response of the heat-release process to flow perturbations depends nonlinearly on the amplitude of the disturbance. For example, a small perturbation in pressure may cause the rate of heat release to fluctuate slightly. However, pressure perturbations of sufficiently large amplitudes may cause the combustion process to temporally extinguish. The large-amplitude heat-release oscillations induced by

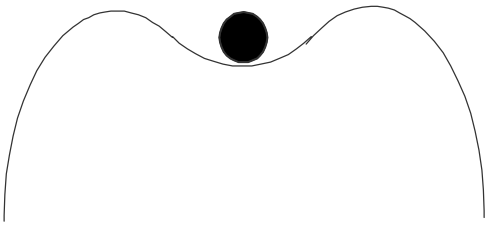

Fig. 1.15 A simple example of a nonlinearly unstable state.

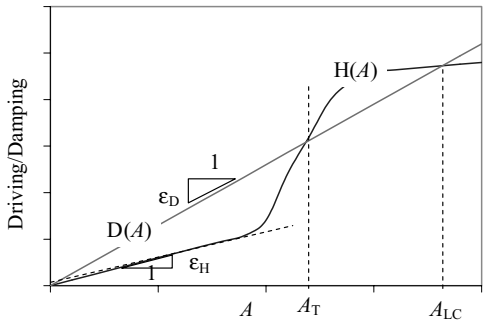

Fig. 1.16 Hypothetical dependence of the acoustic driving, H(A), and damping, D(A), processes on amplitude A, that produce the triggering of instabilities.

these transient extinguishment and reignition processes may cause instability in the combustor. Rockets are prone to nonlinear triggering for reasons that are not fully understood.

Although large-amplitude disturbances are generally required to initiate unstable oscillations in nonlinearly unstable systems,[26] a system may be nonlinearly unstable at low-amplitude disturbances that are of the order of the background-noise level. This scenario is somewhat analogous to the hydrodynamic stability of a laminar Poiseuille flow,[27] which is linearly stable but becomes increasingly susceptible to destabilization by nonlinear mechanisms with increasing Reynolds numbers. It is important to recognize this point because it indicates that it is difficult to determine from measured data whether the combustor is linearly or nonlinearly unstable.

Figure 1.16 provides an example of the amplitude dependences of H(A) and D(A) that produce the behavior discussed earlier. In this case, the system has three equilibrium points at which the driving and damping curves intersect. Specifically, the damping exceeds the driving when $A < A_T$, indicating that $A = 0$ is a stable fixed point, because all disturbances in the range $0 < A < A_T$ decay to $A = 0$. The next equilibrium amplitude at which the driving and damping curves intersect is at the triggering amplitude $A = A_T$. This amplitude is an unstable equilibrium point because any disturbance that shifts the system from this point continues to increase in time. The third equilibrium point, $A = A_{LC}$, is a stable limit cycle. Thus, in such a system all disturbances with amplitudes $A < A_T$ return to the stable solution $A = 0$ and disturbances with amplitudes $A > A_T$ grow until their amplitude attains the value $A = A_{LC}$. Consequently, two stable solutions exist for this operating condition. The one observed at any point in time will depend on the history of the system. This triggering behavior is probably not limited to rockets. As discussed in Chapter 12, nonlinearities in the response of premixed flames to acoustic oscillations result in H(A) curves resembling those shown in both Fig. 1.12 and Fig. 1.16; thus, both subcritical and supercritical bifurcations should be expected in gas turbines as well.

A typical bifurcation diagram for this type of system is shown in Fig. 1.17. It shows that for $\varepsilon_h < \varepsilon_d$, the $A = 0$ and $A = A_{LC}$ are stable solutions, as noted

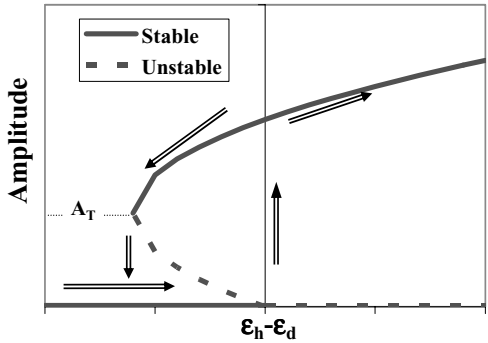

Fig. 1.17 An example of a subcritical bifurcation.

before. For $\varepsilon_h > \varepsilon_d$, the $A = 0$ solution becomes unstable (as indicated by the solid line becoming dashed), and only a single stable solution is present. In this case, if a system parameter is monotonically increased to change the sign of $\varepsilon_h - \varepsilon_d$ from a negative to a positive value, the system's amplitude will jump discontinuously from $A = 0$ to $A = A_{LC}$ at $\varepsilon_h - \varepsilon_d = 0$. Hysteresis is also present in the system, because if the system parameter is subsequently decreased, the system's amplitude decreases as it follows the stable branch on top, even for a range of $\varepsilon_h < \varepsilon_d$ values, before it discontinuously "jumps" to the $A = 0$ solution. Experimental data exhibiting such behavior are presented in Fig. 1.18.[25]

B. Other Characteristics of Limit-Cycle Oscillations

The preceding discussion has shown that both linear and nonlinear combustion and acoustic processes control the dynamics of unstable combustors. In general, linear processes control the balance between driving and damping processes for small-amplitude disturbances and, thus, determine the conditions under which spontaneous self-excited instabilities occur. On the other hand, nonlinear combustor processes control the dynamics of finite amplitude oscillations.

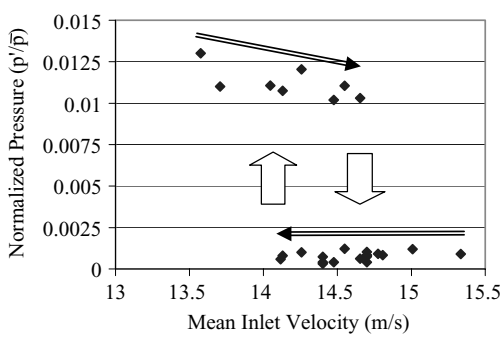

Fig. 1.18 Experimental data showing evidence of a subcritical bifurcation. Taken from Lieuwen.[25]

Nonlinear processes are also responsible for two other phenomena observed in unstable combustors: generation of harmonics and changes in the mean value of certain system properties. These phenomena can be understood by considering the response of a nonlinear system to harmonic forcing. Consider, for example, the gas-dynamic nonlinearity generated by the isentropic relationship between pressure and density; that is,

$$\frac{p(t)}{\bar{p}} = \left(\frac{\rho(t)}{\bar{\rho}}\right)^{\gamma} \tag{1.3}$$

Assume that the density oscillates harmonically with an amplitude A and angular frequency ω, that is,

$$\frac{\rho'(t)}{\bar{\rho}} = A \sin \omega t \tag{1.4}$$

Substitute this expression into Eq. (1.3) and expand the resulting expression in a Taylor series about $A = 0$ to obtain the following expression for the pressure oscillations:

$$\begin{aligned}\frac{p'(t)}{\gamma \bar{p}} = {} & \frac{(\gamma-1)}{4}A^2 + \left(A + \frac{(\gamma-1)(\gamma-2)}{8}A^3\right)\sin \omega t - \frac{(\gamma-1)}{4}A^2 \cos 2\omega t \\ & - \frac{(\gamma-1)(\gamma-2)}{24}A^3 \sin 3\omega t + \cdots \end{aligned} \tag{1.5}$$

Equation (1.5) shows that the expansion for the pressure consists of a linear term that is proportional to A (the second term on the right), and a collection of nonlinear terms that are proportional to higher powers of the amplitude A. The latter includes a time-*independent* term (first term on the right) that is proportional to A^2 and represents the change in the mean pressure caused by the purely oscillatory density disturbance. Furthermore, the expansion includes harmonics that are proportional to A^2, oscillating at twice the disturbance frequency, and terms proportional to A^3, oscillating with frequencies of 3ω and ω. Such harmonics of the unstable combustor mode are routinely observed.

Changes in mean pressure of up to several hundred pounds per square inch have been observed during very-large-amplitude instabilities in solid rockets. On the other hand, instabilities in gas turbines generally have much smaller magnitudes, and no reports have been published of significant shifts in mean pressure because of these instabilities. However, changes in mean flame location and length are routinely observed.[28,29] These changes, in turn, can have an impact on the static pressure distribution in the combustor because it depends on the distribution of the combustion-process heat release.

C. Causes of Nonlinearities

Numerous processes that drive and damp oscillations in unstable combustors depend nonlinearly on the instability amplitude. These nonlinearities may be caused

by flow and combustion processes that occur either within the combustor volume or at its boundaries.

Gas-dynamic nonlinearities within the combustor volume are introduced by processes described by nonlinear terms in the Navier–Stokes and energy equations, for example, convective terms such as $\rho\vec{u} \cdot \nabla\vec{u}$ in the Navier–Stokes equation or the previously discussed nonlinear pressure–density relationship. Such terms, in general, become significant when the amplitudes of the fluctuating pressure, density, or velocity become on the order of the mean pressure, density, or speed of sound, respectively. Consequently, these terms are generally not important when the relative amplitudes of the acoustic disturbances are low (e.g., $p'/\bar{p} < 10\%$). On the other hand, when very-large-amplitude oscillations are encountered, these nonlinear processes strongly affect the characteristics of the instabilities in these systems. Examples of the treatments of these nonlinear processes can be found in, for example, the works of Zinn and coworkers,[30] Culick and coworkers,[31,32] and Yang.[33]

Combustion-process nonlinearities are introduced by the nonlinear dependence of the heat-release oscillations on the disturbance amplitude. Additionally, flow oscillations may control the response of the combustion process, resulting (in these cases) in nonlinearities that become important when the ratio $u'/\bar{u} \sim O(1)$.[25] Consequently, in these cases, the relevant velocity scale that determines when nonlinearities are important is the mean velocity and not the sound speed (as in the gas-dynamic nonlinearities). In fact, it has been shown that such combustion-process nonlinearities play a key role in the stability of lean premixed combustion systems.[34] Further discussion of these nonlinearities can be found in Chapters 12 and 13.

The nonlinearities in processes that occur at or near the combustor boundaries also affect the combustor dynamics as they are introduced into the analysis of the problem through nonlinear boundary conditions. Such nonlinearities are caused by, for example, flow separation at sharp edges or rapid expansions, which, as discussed earlier, cause stagnation-pressure losses and a corresponding transfer of acoustic energy into vorticity. The resulting nonlinear damping is proportional to $\rho|\vec{u}|^2$, resulting in an unsteady damping process that is proportional to $\bar{\rho}(|\vec{u}'|^2 + 2|\vec{u}'| \cdot |\bar{\vec{u}}|) + \rho'(|\bar{\vec{u}}|^2 + |\vec{u}'|^2 + 2|\vec{u}'| \cdot |\bar{\vec{u}}|)$.[35] This expression indicates that the presence of mean flow introduces linear and nonlinear damping terms and that the damping is proportional to the square of the magnitude of the velocity amplitude $|\vec{u}'|^2$ in the absence of mean flow (i.e., $|\bar{\vec{u}}| = 0$). Such nonlinear damping has been well characterized in experiments (see Ref. 35, for example).

Also, wave reflection and transmission processes through choked and unchoked nozzles become amplitude dependent at large amplitudes. The effect of these processes on the instabilities is described by a nonlinear boundary condition.[36]

V. Conclusion

This chapter concludes with a summary of the state of the art of current modeling capabilities of combustion instabilities. From a practical point of view, three basic instability characteristics must be predicted and understood: 1) *frequency* of oscillations, 2) *conditions* under which the oscillations occur, and 3) their final, limit-cycle *amplitude*, which are listed in increasing order of predictive difficulty.

Specifically, the frequency and mode shape are easier to predict because they generally only require an understanding of the system's linear dynamics, whereas determination of the limit-cycle amplitude requires an understanding of nonlinear system characteristics.

Consider first the prediction of instability frequencies. Although some fundamental problems remain, such as analytical descriptions of the combustor's acoustic boundary conditions, prediction of instability frequencies is a relatively mature area. In fact, capabilities for modeling the acoustics of combustors are reasonably well developed, as described in Chapter 13, in this volume and also, for example, in recent proceedings of the American Society of Mechanical Engineers Turbo Expo conferences or in Munjal's book.[37] These descriptions indicate that the frequencies and mode shapes of the excited instabilities can often be predicted with good accuracy. In general, accurate predictions of these properties simply require knowledge of the geometric characteristics of the system and average temperature distributions. Although unsteady heat-release effects, which are much harder to predict, have some impact on instability frequencies, this impact is often small.

Predicting the conditions under which instabilities occur (the second issue) is considerably more difficult than predicting acoustic mode shapes and frequencies, because it requires knowledge and modeling of the interactions of flow and mixture disturbances with flames and damping processes. In recent years, the focus of much of the research in this area has been on developing these understanding and modeling capabilities. Examples of the findings of some of these studies that investigated, for example, acoustic wave–flame interactions and other mechanisms that can drive these instabilities are discussed Chapters 9 and 12. Much progress has been made in these areas to date and many gas-turbine manufacturers have reported success in predicting instability frequencies and mode shapes, and the conditions under which they occur, as discussed in the subsequent case-study chapters in this book. Furthermore, even in cases in which predictive capabilities do not exist, reasonable understanding of qualitative combustor stability trends can often be obtained through analysis of combustor data after they have been obtained; for example, rational correlations of the data can be developed.

Note that direct experimental verification of linear combustor dynamics models and mechanisms of instability is difficult, because instabilities, in general, are experimentally studied under limit-cycle conditions (with the notable exception of Ref. 38). For example, consider the role of vortex shedding, which is thought to be an important instability mechanism in gas turbines. The mere observation of vortex shedding at the instability frequency under limit-cycle conditions does not necessarily imply that it is also responsible for initiating the instability. Vortex shedding can be forced at the frequency of an external excitation when the forcing occurs at sufficiently large amplitude, even if this frequency does not coincide with the natural shedding frequency.[39] Consequently, flow oscillations may be excited by some other instability mechanism, which subsequently forces the shedding of vortices at the instability frequency.

In contrast to the prediction of the instability characteristics and the conditions under which it spontaneously occurs, which can be determined with linear combustor models, the prediction of the limit-cycle amplitude of the instability and the conditions under which large-amplitude disturbances destabilize a linearly

stable system (triggering) requires solution of nonlinear models of the combustor processes. Amplitude-prediction capabilities are critical because it would tell engineers whether an instability can be tolerated in the system under consideration if its amplitude is not too large, or whether they need to take steps to decrease its magnitude to prevent damage to the combustor. Such capabilities require understanding of the flame and gas-dynamic response to large-amplitude disturbances and the ability to model these nonlinear phenomena. Progress in this area has largely been limited to simple, laminar flames and little is known, even in a qualitative sense, about the key parameters controlling nonlinear flame dynamics. As such, instability-amplitude prediction capabilities remain a key challenge for future work.

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